

Pizza Sales Data Analysis Report

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Abstract

This report presents an in-depth analysis of pizza sales data, comprising 48,620 transactions with 12 key attributes including order details, pizza variants, quantities, and revenue. The analysis aims to uncover sales trends, customer purchase behaviour, and product performance insights using SQL queries on a MySQL database. These insights enable strategic business decisions to optimize sales and improve customer satisfaction.

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Introduction

Pizza sales data offer rich insights into customer preferences, effective product offerings, and temporal sales variations. Analysing this data helps identify best-selling products, optimal pricing strategies, and peak sales periods. This project imports the pizza sales dataset into MySQL Workbench and applies various SQL queries to explore and summarize the data. Results include total revenue, sales by pizza size and category, monthly and daily trends, and ranking of products by quantity and revenue.

Exploratory Data Analysis (EDA)

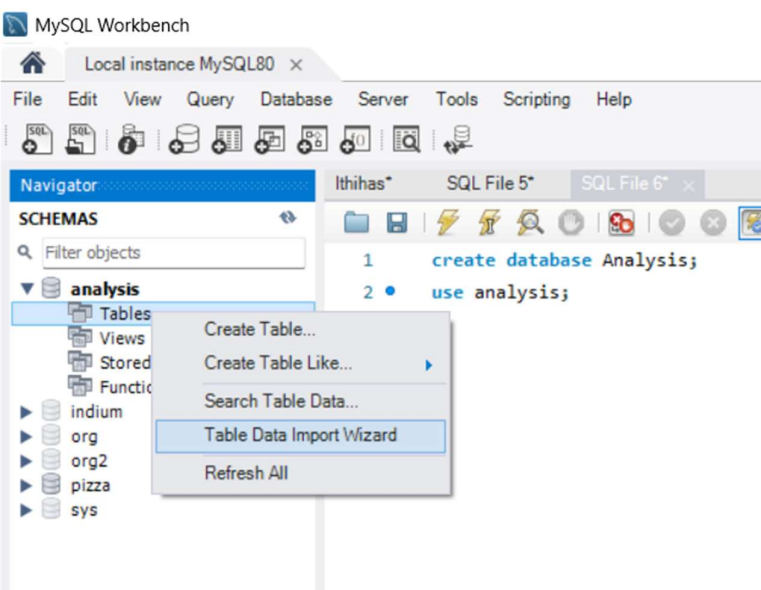
The structure and content of the dataset

id	pizza_id	order_id	pizza_name_id	quantity	order_date	order_time	unit_price	total_price	pizza_size	pizza_category	pizza_ingredients	pizza_name
1	2	1	1	havalian_m	1	01-01-2015	11:38:36	13.25	13.25 M	Classic	Sliced Ham, Pineapple, Mozzarella Cheese	The Hawaiian Pizza
2	2	2	2	classic_dlx_m	1	01-01-2015	11:57:40	16	16 M	Classic	Pepperoni, Mushrooms, Red Onions, Red Peppers, Bacon	The Classic Deluxe Pizza
3	3	2	3	five_cheese_l	1	01-01-2015	11:57:40	18.5	18.5 L	Veggie	Mozzarella Cheese, Provolone Cheese, Smoked Gouda Cheese, Romano Cheese, Blue Cheese, G	The Five Cheese Pizza
4	3	2	4	ital_supr_l	1	01-01-2015	11:57:40	20.75	20.75 L	Supreme	Calabrese Salami, Capocollo, Tomatoes, Red Onions, Green Olives, Garlic	The Italian Supreme Pizza
5	2	1	5	metaxiana_m	1	01-01-2015	11:57:40	16	16 M	Veggie	Tomatoes, Red Peppers, Jalapeno Peppers, Red Onions, Cilantro, Corn, Chipotle Sauce, Garlic	The Mexicana Pizza
6	2	1	6	chick_pizze_l	1	01-01-2015	11:57:40	20.75	20.75 L	Chicken	Chicken, Pineapple, Tomatoes, Red Peppers, Thai Sweet Chili Sauce	The Chicken Pizza
7	3	1	7	ital_supr_m	1	01-01-2015	12:12:28	16.5	16.5 M	Supreme	Calabrese Salami, Capocollo, Tomatoes, Red Onions, Green Olives, Garlic	The Italian Supreme Pizza
8	3	1	8	prsc_argula_l	1	01-01-2015	12:12:28	20.75	20.75 L	Supreme	Prosciutto di San Daniele, Arugula, Mozzarella Cheese	The Prosciutto and Arugula Pizza

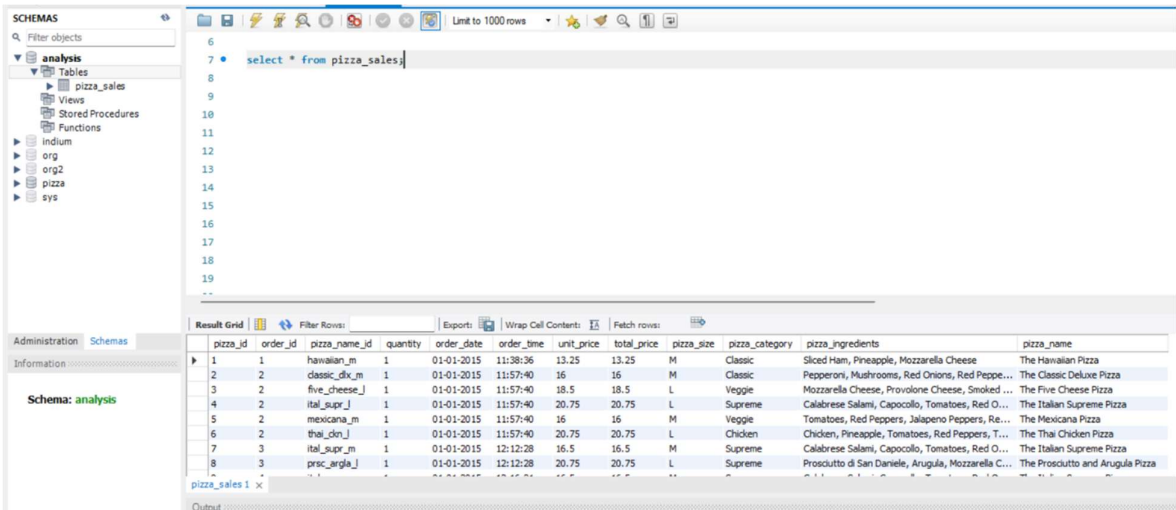
Source: <https://www.kaggle.com/code/mdismielhossenabir/pizza-sales-dataset>

The dataset contains 48,620 rows and 12 columns, including pizza identification, quantity, order timing, pricing, size, category, ingredients, and names, providing a comprehensive view for sales analysis.

Importing data into MySQL

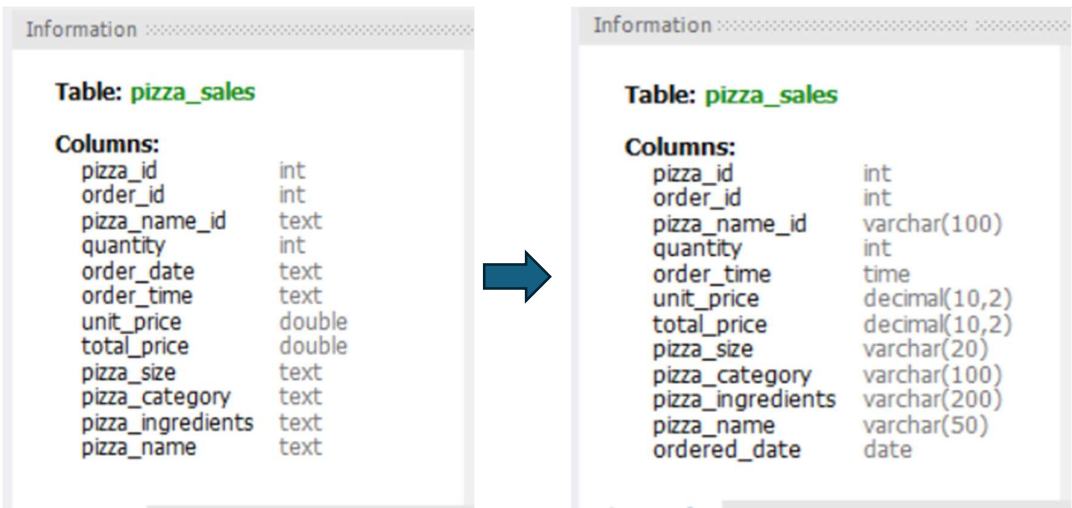


The "analysis" database schema was created and selected to organize data and facilitate query execution. Data import was conducted using MySQL Workbench's Table Data Import Wizard into 'Analysis' database by creating 'pizza_sales' table, ensuring correct schema mapping.



Verified with a sample select query showing all columns correctly imported, ready for analytical querying.

Changing Data Types



Business Questions

Business Question 1: What is the total revenue from pizza sales?

```
58  -- 1. the sum of the total price of all pizza orders
59 • SELECT SUM(total_price) as Total_sales
60   from pizza_sales;
--
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Total_sales			
817860.05			

Aggregate sales amount was calculated, revealing total revenue of 817,860.05 over the data period.

Business Question 2: How many pizzas were sold?

```
68  -- 3. total pizza sold
69 • select SUM(quantity) as pizza_sold
70   from pizza_sales;
--
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
pizza_sold			
49574			

Summing quantities gives a total of 49,574 pizzas sold, indicating product volume.

Business Question 3: What is the average spending per order?

```
63  -- 2. avg spend per order
64 • select SUM(total_price)/COUNT(distinct order_id) as Avg_price_per_order
65   from pizza_sales;
--
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Avg_price_per_order			
38.307262			

Mean aggregate price per distinct order calculated as 38.31, key for revenue and pricing insights.

Business Question 4: How many unique orders were placed?

```
72
73  -- 4. total number of orders
74 • select count(distinct order_id) as total_orders
75    from pizza_sales;
--
```



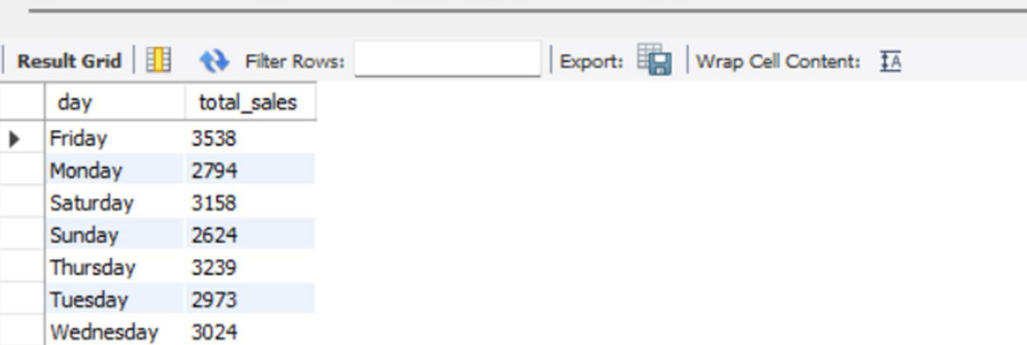
Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	total_orders
▶	21350

Unique orders count stands at 21,350, reflecting business transaction volume.

Business Question 5: How do sales distribute by day of the week?

```
86  -- 1.daywise sales
87 • select date_format(ordered_date, '%W') as day,
88          count(distinct order_id) as total_sales
89    from pizza_sales group by day;
--
```



Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	day	total_sales
▶	Friday	3538
	Monday	2794
	Saturday	3158
	Sunday	2624
	Thursday	3239
	Tuesday	2973
	Wednesday	3024

Highest sales observed on Fridays, lowest on Sundays, informing staffing and promotional plans.

Business Question 6: What are monthly sales trends?

```
93 -- 2. monthwise sales
94 • SELECT DATE_FORMAT(ordered_date, '%M') AS month,
95         count(distinct order_id) as total_sales
96 from pizza_sales
97 group by month;
98
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
month	total_sales		
April	1799		
August	1841		
December	1680		
February	1685		
January	1845		
July	1935		
June	1773		
March	1840		
May	1853		
November	1792		
October	1646		
September	1661		

Sales volumes fluctuate, with peaks in April and January, useful for seasonal inventory planning.

Business Question 7: What is the percentage of sales by pizza size?

```
99 -- 3. % of sales by pizza size
100 • select pizza_size,
101         CAST(sum(total_price)*100/(select sum(total_price) from pizza_sales) AS decimal(10,2)) as PTS
102 from pizza_sales
103 group by pizza_size;
104
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
pizza_size	PTS		
M	30.49		
L	45.89		
S	21.77		
XL	1.72		
XXL	0.12		

'Large' pizza accounts for the highest sales percentage (45.89%), indicating size popularity.

Business Question 8: How many pizzas sold by category?

```
---
106      -- 4. total pizzas sold by pizza category
107 •    select pizza_category,
108           sum(quantity)
109      from pizza_sales
110      group by 1;
111
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
pizza_category		sum(quantity)	
▶	Classic	14888	
	Veggie	11649	
	Supreme	11987	
	Chicken	11050	

'Classic' category leads in quantity sold, followed by 'Veggie', 'Supreme', and 'Chicken'.

Business Question 9: Who are the top 5 best sellers by quantity?

```
113      -- 5.1 top 5 best sellers by qty
114 •    select pizza_name,
115           sum(quantity) as sold_qty
116      from pizza_sales
117      group by pizza_name
118      order by sold_qty desc
119      limit 5;
120
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
pizza_name		sold_qty		
▶	The Classic Deluxe Pizza	2453		
	The Barbecue Chicken Pizza	2432		
	The Hawaiian Pizza	2422		
	The Pepperoni Pizza	2418		
	The Thai Chicken Pizza	2371		

Classic Deluxe and Barbecue Chicken are among the highest selling pizzas by volume.

Business Question 10: Who are the top 5 best sellers by revenue?

```
121 -- 5.2 top 5 best sellers by revenue
122 • select pizza_name,
123         sum(total_price) as revenue
124 from pizza_sales
125 group by pizza_name
126 order by revenue desc
127 limit 5;
128
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
pizza_name		revenue		
▶	The Thai Chicken Pizza	43434.25		
	The Barbecue Chicken Pizza	42768.00		
	The California Chicken Pizza	41409.50		
	The Classic Deluxe Pizza	38180.50		
	The Spicy Italian Pizza	34831.25		

Revenue leaders include Thai Chicken and Barbecue Chicken, guiding focus for high-revenue products.

Business Question 11: Who are the top 5 pizzas by order count?

```
129 -- 5.3 top 5 best sellers by orders
130 • select pizza_name,
131         count(distinct order_id) as orders
132 from pizza_sales
133 group by 1
134 order by orders desc
135 limit 5;
136
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
pizza_name		orders		
▶	The Classic Deluxe Pizza	2329		
	The Hawaiian Pizza	2280		
	The Pepperoni Pizza	2278		
	The Barbecue Chicken Pizza	2273		
	The Thai Chicken Pizza	2225		

Classic Deluxe Pizza topped order counts, indicating strong customer preference.

Business Question 12: Who are the bottom 5 pizzas by quantity?

```
137 -- 6.1 bottom 5 sellers by qty
138 • select pizza_name,
139         sum(quantity) as sold_qty
140 from pizza_sales
141 group by pizza_name
142 order by sold_qty
143 limit 5 ;
144
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
pizza_name		sold_qty		
▶	The Brie Carre Pizza	490		
	The Mediterranean Pizza	934		
	The Calabrese Pizza	937		
	The Spinach Supreme Pizza	950		
	The Soppressata Pizza	961		

Identified lower demand pizzas to consider for menu optimization.

Business Question 13: Who are the bottom 5 pizzas by revenue?

```
145 -- 6.2 bottom 5 sellers by revenue
146 • select pizza_name,
147         sum(total_price) as revenue
148 from pizza_sales
149 group by pizza_name
150 order by revenue
151 limit 5;
152
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
pizza_name		revenue		
▶	The Brie Carre Pizza	11588.50		
	The Green Garden Pizza	13955.75		
	The Spinach Supreme Pizza	15277.75		
	The Mediterranean Pizza	15360.50		
	The Spinach Pesto Pizza	15596.00		

Low revenue pizzas suggest candidates for marketing or discontinuation review.

Business Question 14: Who are the bottom 5 pizzas by orders?

```
153  -- 6.3 bottom 5 sellers by orders
154  •  select pizza_name,
155         count(distinct order_id) as orders
156  from pizza_sales
157  group by 1
158  order by orders
159  limit 5;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Fetch rows:

	pizza_name	orders
▶	The Brie Carre Pizza	480
	The Mediterranean Pizza	912
	The Calabrese Pizza	918
	The Spinach Supreme Pizza	918
	The Chicken Pesto Pizza	938

This highlights low order-frequency pizzas for focus.

Conclusion

The analysis provides an extensive view of pizza sales dynamics, customer preferences, and business performance metrics. Insights on best-selling products, sales seasonality, and customer ordering patterns equip stakeholders to tailor marketing campaigns, adjust inventory, and optimize revenues. Future work includes deeper predictive analytics and customer segmentation for enhanced strategic planning.